Canopy (Dense Jungle / Recon & Autonomous Search)

Executive Summary

Canopy is an ultra-dense **tropical jungle reconnaissance battlefield** built for **autonomous search, stealth movement, sensor-limited engagements, and foliage-driven concealment**. The terrain features towering canopy layers, tangled undergrowth, vine-choked ravines, mossy cliffs, hidden trails, abandoned outposts, and river-cut gorges. The map emphasizes Obsidian Protocol's autonomous recon logic, sensor-fusion under heavy occlusion, comms-relay warfare, and logistics under extreme vegetation density. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of deep-jungle terrain. Key features include multi-layer canopy concealment, autonomous search corridors, unpredictable sightlines, and stealth-heavy engagements.

Design Goals & Gameplay

Dense-Jungle Reconnaissance

Canopy prioritizes scouting, tracking, and stealth infiltration. Vegetation density creates natural blind zones, forcing reliance on autonomous recon units and sensor triangulation.

Autonomous Search & Pathfinding

Units navigate canopy shadows, narrow trails, root-choked paths, and vine-covered ravines. Autonomous logic selects routes based on concealment, sensor clarity, and terrain stability.

Concealment & Ambush Warfare

Foliage, canopy layers, and humidity fog create stealth pockets. Ambushes, flanking, and sudden close-range engagements dominate.

Sensor & Information Warfare

Humidity, fog, vegetation clutter, thermal distortion, and canopy occlusion degrade sensors. Relay towers on cliffs restore clarity. Destroying relays creates blind pockets across the jungle.

Logistics & Resources

Fuel caches, abandoned outposts, and river docks serve as resupply nodes. Convoys must navigate narrow jungle trails and avoid ambush-prone canopy corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between stealth infantry, recon drones, or heavy jungle-adapted armor. Blue deploys from the southwest river-trail camp; Red deploys from the northeast cliffside recon tower.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² dense-jungle zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (River-Trail Recon Camp) — (0,0), 150×150 m Boats, crates, recon drones, jungle staging.
Red HQ (Cliffside Recon Tower) — (1000,1000), 150×150 m Elevated bunker with radar mast, thermal relays, and canopy-overwatch.
Central Canopy Basin — (500,450), 300×300 m Dense canopy, fog pockets, unpredictable sightlines, stealth-heavy combat.
Abandoned Outpost Cluster — (250,800), 120×120 m Ruins, generators, interior flanking routes.
Ridge-Top Canopy Network — (800,300), 200×200 m High-ground overwatch, tree-top concealment, stealth movement.
Recon Corridor — (0,450 → 400,900) Heavy vegetation zone; extreme sensor degradation.
Hidden Trail Route — (300,0 → 450,350) Safer but slow; root hazards and vegetation cover.
Cliff Relay A — (650,150), +240 m Long-range sensor vantage point.
Cliff Relay B — (150,650), +260 m Comms relay hub and recon-monitoring node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Recon Corridor

Severe visibility loss; ideal for stealth ambushes and sensor deception.

Ridge-Top Canopy Network

Interior flanking routes; unpredictable cover due to foliage.

Canopy Basin

Long-range engagements broken by vegetation; humidity reduces accuracy.

Hidden Trail Route

Predictable but safe; vulnerable to overwatch from ridges.

Sightlines

Cliff Relay A provides long-range sensor coverage (~300 m). Fog and canopy shadows create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW river-trail camp + recon staging.
Red Deployment: NE cliffside recon tower + thermal defenses.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Canopy layers, cliffs, rivers, mud trails, abandoned outposts.

Vegetation

Dense trees, vines, ferns, underbrush, mossy cliffs.

Buildings & Props

Military: Radar dishes, thermal relays, bunkers. Rural: Huts, bridges, wooden structures. Logistics: Trucks, crates, fuel drums, boats, drones.

Effects

Fog, humidity haze, rain, canopy shadows, mud splashes.

Audio

Insects, rainfall, distant machinery, river flow, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Jungle Te	3	200k	600k
Rural/Ind	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048^2 for terrain/buildings, 1024^2 for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out canopy basin, ridges, outposts, AR anchor test.
2. **Core Systems — Month 2** NavMesh, jungle autonomy behaviors, recon logic.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (fog, humidity haze, canopy shadows).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify jungle navigation, recon-aware pathfinding, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under canopy occlusion.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, recon logic, deployment budget enforcement, anchor drift, sensor accuracy.